

Microeconomics

Lecture 9: Cost Minimization and Optimal Input Choice

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What we know so far

- technology, isoquants, $MRTS = MPL/MPK$, returns to scale
- long-run costs: TC , AC , MC , their shape and their relationships

Next step

How the long-run cost function *arises*, i.e., given Q , which mix (L, K) costs the least?

Learning objectives

By the end of this lecture you will be able to:

1. Set up the minimization problem: $\min wL + rK$ s.t. $f(L, K) \geq Q$
2. Construct and interpret **isocost** lines
3. Derive the **tangency condition**: $MRTS = w/r$
4. Solve the problem with a Cobb–Douglas technology
5. Do **comparative statics**: effects of changes in w or r on (L^*, K^*)
6. Link optimal input choice to real business decisions

Symmetry with the consumer

The consumer *maximizes utility* given the budget.

The firm *minimizes cost* given the production constraint.

Same mathematical structure, reversed direction.

The cost-minimization problem (LR)

$$\min_{L,K} C = wL + rK \quad \text{s.t.} \quad f(L, K) \geq Q$$

- w : wage (cost per unit of labor)
- r : cost of capital (leasing fee, required rate of return)
- The constraint is the isoquant $f(L, K) = Q$

Geometric objective: find the point on the isoquant that lies on the lowest possible isocost.

Example: a logistics firm

It must deliver 10,000 parcels per day ($Q = 10,000$). It can do so with many couriers and few vans, or with many automated vans and few couriers. The optimal mix depends on w (labor cost) and r (van leasing cost).

Isocost lines

The isocost is the set of combinations (L, K) that cost the same:

$$C = wL + rK \quad \Rightarrow \quad K = \frac{C}{r} - \frac{w}{r}L$$

- K -intercept: C/r —how many units of K you can buy with the whole budget
- L -intercept: C/w —how many units of L with the whole budget
- **Slope**: $-w/r$ —the relative price of labor in terms of capital

Shifts of the isocost:

- Higher isocost (parallel, farther from the origin) $\Rightarrow$ greater expenditure
- If $w \uparrow$: the isocost becomes steeper (labor is relatively more expensive)
- If $r \uparrow$: the isocost becomes flatter

Exercise: Isocost

Given $w = 12$, $r = 24$, budget $C = 240$.

1. Write the isocost equation in the form $K = \dots$
2. Find the intercepts on L and on K .
3. What is the slope?
4. If w rises to 18 (with r and C unchanged), how does the isocost change? Sketch it.

Solution

$$240 = 12L + 24K \Rightarrow K = \frac{240}{24} - \frac{12}{24}L = 10 - \frac{1}{2}L$$

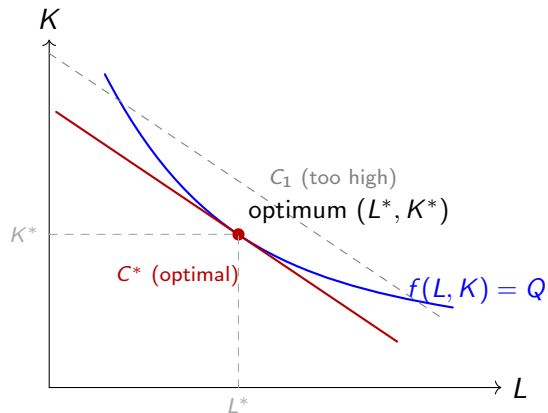
Intercept on K : $L = 0 \Rightarrow K = 10$

Intercept on L : $K = 0 \Rightarrow L = 20$

$$\text{Slope} = -\frac{w}{r} = -\frac{12}{24} = -\frac{1}{2}$$

If $w = 18$: $K = 10 - \frac{18}{24}L = 10 - \frac{3}{4}L$. **The isocost pivots inward on the L axis:** the L -intercept falls to $240/18 \approx 13.3$, and the isocost becomes steeper (slope $-3/4$).

The optimality condition: isoquant + isocost



Tangency condition: $MRTS = w/r$

At an interior optimum, the isoquant and the isocost are **tangent**:

$$MRTS_{LK} = \frac{MPL}{MPK} = \frac{w}{r}$$

Equivalent form (marginal product per euro spent):

$$\frac{MPL}{w} = \frac{MPK}{r}$$

Intuition

If $\frac{MPL}{w} > \frac{MPK}{r}$: one more euro spent on labor produces more output than one more euro spent on capital. It pays to shift spending toward L (and reduce K) until the returns per euro are equalized.

Exercise: Optimality condition

At a point on the isoquant, a firm observes:

$$MPL = 8, \quad MPK = 4, \quad w = 16, \quad r = 24$$

1. Compute $\frac{MPL}{MPK}$ (MRTS) and $\frac{w}{r}$.
2. Are we at the optimum?
3. If we are not at the optimum, in which direction should the firm move? (more L and less K , or vice versa?)
4. What happens to the marginal products as we move toward the optimum?

Solution

$$MRTS = \frac{MPL}{MPK} = \frac{8}{4} = 2, \quad \frac{w}{r} = \frac{16}{24} = \frac{2}{3}$$

We are not at the optimum: $MRTS = 2 > 2/3 = w/r$.

Equivalently:

$$\frac{MPL}{w} = \frac{8}{16} = 0.5 > \frac{MPK}{r} = \frac{4}{24} \approx 0.167$$

Labor yields more per euro spent $\Rightarrow$ the firm should **increase L and reduce K** .

As the firm moves: MPL tends to fall (diminishing returns to L), MPK tends to rise (less K used), until $MPL/w = MPK/r$.

Exercise: Minimization with the Lagrangian

Technology: $Q = L^{0.5}K^{0.5}$. Prices: $w = 4$, $r = 9$. Target: $Q = 6$.

Set up and solve the problem with the **Lagrangian**:

$$\mathcal{L}(L, K, \lambda) = wL + rK - \lambda[f(L, K) - Q]$$

1. Write $\mathcal{L}$ explicitly with the numerical values.
2. Derive the three first-order conditions.
3. Find L^* , K^* and the value of the multiplier λ^* .
4. Give an economic interpretation of λ^* .

Solution

The Lagrangian:

$$\mathcal{L} = 4L + 9K - \lambda [L^{0.5}K^{0.5} - 6]$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial L} = 4 - \lambda \cdot 0.5L^{-0.5}K^{0.5} = 0$$

$$\frac{\partial \mathcal{L}}{\partial K} = 9 - \lambda \cdot 0.5L^{0.5}K^{-0.5} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -[L^{0.5}K^{0.5} - 6] = 0$$

Dividing the first by the second:

$$\frac{4}{9} = \frac{K^{0.5}L^{-0.5}}{L^{0.5}K^{-0.5}} = \frac{K}{L} \quad \Rightarrow \quad K = \frac{4}{9}L$$

We recover exactly the condition $MRTS = w/r$: geometry and calculus say the same thing.

Solution

Output constraint:

$$L^{0.5} \left(\frac{4}{9} L \right)^{0.5} = 6 \Rightarrow L \cdot \left(\frac{4}{9} \right)^{0.5} = 6 \Rightarrow L^* = \frac{6}{\frac{2}{3}} = 9$$

$$K^* = \frac{4}{9} \cdot 9 = 4$$

Minimum cost:

$$C^* = 4 \cdot 9 + 9 \cdot 4 = 36 + 36 = 72$$

Multiplier λ^* (from the first FOC):

$$\lambda^* = \frac{4}{0.5 \cdot L^{*-0.5} K^{*0.5}} = \frac{4}{0.5 \cdot 9^{-0.5} \cdot 4^{0.5}} = \frac{4}{0.5 \cdot \frac{1}{3} \cdot 2} = \frac{4}{1/3} = 12$$

Interpretation of λ^*

$\lambda^* = 12$ is the **long-run marginal cost**: producing one additional unit beyond $Q = 6$ costs exactly 12 more. The multiplier measures how much the minimum cost increases at the margin as the output target rises.

Cobb–Douglas technology

For $Q = AL^\alpha K^\beta$, the tangency condition becomes:

$$MRTS = \frac{MPL}{MPK} = \frac{\alpha}{\beta} \cdot \frac{K}{L} = \frac{w}{r}$$
$$\Rightarrow \frac{K}{L} = \frac{\beta}{\alpha} \cdot \frac{w}{r} \Rightarrow K = \frac{\beta}{\alpha} \cdot \frac{w}{r} \cdot L$$

Complete procedure:

1. Obtain the ratio K/L from the tangency condition
2. Substitute into the constraint $f(L, K) = Q$ to find L^*
3. Obtain K^* from the ratio
4. Compute $C^* = wL^* + rK^*$

Exercise: Complete minimization

Technology: $Q = L^{0.6}K^{0.4}$

Prices: $w = 10$, $r = 20$. Target: $Q = 50$.

1. Find the optimal ratio K/L .
2. Find L^* and K^* .
3. Compute the minimum cost $C^* = wL^* + rK^*$.
4. How does the ratio K/L change if w rises to 15?

Solution

Step 1—Optimal ratio:

$$\frac{K}{L} = \frac{\beta}{\alpha} \cdot \frac{w}{r} = \frac{0.4}{0.6} \cdot \frac{10}{20} = \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}$$
$$\Rightarrow K = \frac{L}{3}$$

Step 2—Output constraint (note: $\alpha + \beta = 1$):

$$50 = L^{0.6} \left(\frac{L}{3} \right)^{0.4} = L^{0.6} \cdot L^{0.4} \cdot 3^{-0.4} = L \cdot 3^{-0.4}$$

$$L^* = 50 \cdot 3^{0.4} \approx 50 \cdot 1.552 \approx 77.6$$

$$K^* = \frac{L^*}{3} = \frac{50 \cdot 3^{0.4}}{3} = 50 \cdot 3^{-0.6} \approx 25.9$$

Solution

Step 3—Minimum cost:

$$C^* = wL^* + rK^* = 10 \cdot 77.6 + 20 \cdot 25.9 = 776 + 518 = 1,294$$

Interpretation

With $r = 20 > w = 10$, capital is relatively expensive: the optimal mix is $K/L = 1/3$, i.e., much more labor than capital. The firm is labor-intensive.

Step 4—If w rises to 15:

$$\frac{K}{L} = \frac{0.4}{0.6} \cdot \frac{15}{20} = \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}$$

The ratio K/L rises from $1/3$ to $1/2$: the firm substitutes capital for labor.

Comparative statics: effects of w and r

From the optimality condition $\frac{K}{L} = \frac{\beta}{\alpha} \cdot \frac{w}{r}$:

Shock	Effect on K/L	Interpretation
$w \uparrow$	$K/L \uparrow$	Replaces L with K (automation)
$w \downarrow$	$K/L \downarrow$	Uses more labor, less capital
$r \uparrow$	$K/L \downarrow$	Capital more expensive, prefers L
$r \downarrow$	$K/L \uparrow$	Investment subsidy $\Rightarrow$ more K

Note: The minimum cost C^* always rises when w or r rise (you pay more for the same output).

Example: Amazon and warehouse automation

Context: Amazon has invested billions in Kiva robots for its fulfillment centers.

Interpretation in terms of w and r :

- Over the last 10 years, wages in the logistics sector in the US (and in Italy) have risen
- $w \uparrow \Rightarrow$ the optimal ratio K/L rises: it pays to replace labor with robots
- At the same time, the cost of robots has fallen ($r \downarrow$) $\Rightarrow$ multiplicative effect

Empirical result

In Amazon warehouses with robots, productivity per m^2 rose by 25% and the cost per unit shipped fell. The model predicts that by 2030 a significant share of picking operations will be automated.

Question: does the same reasoning apply to logistics SMEs in Emilia? Why or why not?

Exercise: Comparative statics

A firm uses technology $Q = L^{0.5}K^{0.5}$, with $w = 20$ and $r = 20$.

1. Find the optimal ratio K/L .
2. What is special here? (Consider the symmetry of the parameters.)
3. Now r falls to 10 thanks to a public investment subsidy. New K/L ?
4. Does the minimum cost of producing $Q = 100$ increase or decrease?

Solution

$$\frac{K}{L} = \frac{0.5}{0.5} \cdot \frac{20}{20} = 1 \Rightarrow K = L$$

Symmetric case: with $\alpha = \beta$ and $w = r$, the firm uses equal quantities of labor and capital.

With $r = 10$:

$$\frac{K}{L} = 1 \cdot \frac{20}{10} = 2 \Rightarrow K = 2L$$

The subsidy makes capital twice as cheap $\Rightarrow$ the firm becomes capital-intensive.

For $Q = 100$: $100 = L^{0.5}K^{0.5} = L^{0.5}(2L)^{0.5} = \sqrt{2} \cdot L \Rightarrow L^* = 100/\sqrt{2} \approx 70.7$, $K^* \approx 141.4$.

$$C^* = 20 \cdot 70.7 + 10 \cdot 141.4 = 1,414 + 1,414 = 2,828$$

Without the subsidy ($r = 20$): $C_{\text{old}}^* = 20 \cdot 100 + 20 \cdot 100 = 4,000$. The subsidy **reduces the minimum cost**.

Example: Energy transition and the choice of K/L

Context: European climate policy raises the cost of traditional energy and provides subsidies for investment in energy efficiency and renewables (e.g. Industry 5.0, the NRRP).

Effects on the minimization problem:

- Energy enters as a third input (E), but we can think of it as part of K
- Subsidies for green investment $\Rightarrow r_{\text{green}} \downarrow$
- Firms that adopt efficient plants lower their LR marginal cost

The Sacmi case

The Sacmi group (machinery for ceramics, packaging, beverage) invested in low-energy plants. Result: while competitors saw their MC rise with the energy crisis, Sacmi kept its LAC more stable.

Exercise: Discussion of policy implications

Scenario: the government introduces a minimum wage that raises w from 10 to 14 in a manufacturing sector, with $r = 20$ unchanged. The technology is $Q = L^{0.7}K^{0.3}$.

1. How does the optimal ratio K/L change?
2. Which firms can adapt easily and which cannot?
3. Does the minimum cost of producing the same Q increase or decrease?
4. Is a firm with high FC (plants already purchased) more or less exposed in the short run?

Solution

$$\left. \frac{K}{L} \right|_{w=10} = \frac{0.3}{0.7} \cdot \frac{10}{20} \approx 0.214 \quad \left. \frac{K}{L} \right|_{w=14} = \frac{0.3}{0.7} \cdot \frac{14}{20} \approx 0.300$$

The ratio K/L rises: the firm tends to substitute capital for labor.

Who adapts: firms with flexible technology, access to credit to invest in automation, sectors in which L and K are good substitutes.

Who suffers: SMEs with fixed plants, labor-intensive sectors with little substitutability (e.g. personal care services).

The minimum cost C^* **rises** (one input is more expensive)—but by less than if there were no possibility of substitution.

With high FC already incurred: in the SR fixed costs are *sunk*; the firm looks only at VC . In the LR it can readjust the composition of its capital.

Exercise: Isocost

A firm has $w = 15$, $r = 30$, budget $C = 300$.

1. Write the isocost equation in the form $K = \dots$
2. Intercepts and slope.
3. If the budget rises to $C = 450$ (with prices unchanged), how does the isocost shift?
4. If instead r falls to 15 (with $C = 300$ and $w = 15$ unchanged), how does it change?

Solution

$$300 = 15L + 30K \Rightarrow K = 10 - \frac{1}{2}L$$

Intercepts: K -int = 10; L -int = 20. Slope = $-1/2$.

$C = 450$: $K = 15 - \frac{1}{2}L$. **Parallel outward shift** (same slope, higher intercepts: $K = 15$, $L = 30$).

$r = 15$: $K = \frac{300}{15} - \frac{15}{15}L = 20 - L$. **Rotation**: the K -intercept rises to 20, the slope becomes -1 (steeper— L and K now have the same price).

Exercise: Optimality condition

A firm is at a point on the isoquant with:

$$MPL = 12, \quad MPK = 6, \quad w = 30, \quad r = 20$$

1. Check whether we are at the optimum using the condition $MRTS = w/r$.
2. Check equivalently with MPL/w vs MPK/r .
3. State the direction of adjustment.
4. What do we expect to happen to MPL and MPK as the firm moves toward the optimum?

Solution

$$MRTS = \frac{12}{6} = 2 \quad \frac{w}{r} = \frac{30}{20} = 1.5$$

$MRTS = 2 > 1.5 = w/r$: we are not at the optimum.

Equivalently:

$$\frac{MPL}{w} = \frac{12}{30} = 0.4 > \frac{MPK}{r} = \frac{6}{20} = 0.3$$

$MPL/w = 0.4 > MPK/r = 0.3$: labor yields more per euro. The firm should **increase L and reduce K** .

Moving toward more L and less K : MPL falls (diminishing returns), MPK rises. The firm converges to the optimum where $MRTS = w/r = 1.5$.

Exercise: Complete minimization

Technology: $Q = 2L^{0.5}K^{0.5}$. Prices: $w = 8$, $r = 2$. Target: $Q = 40$.

1. Find the optimal K/L .
2. Find L^* and K^* .
3. Compute C^* .
4. Comment on the mix (L^*, K^*) : is it labor-intensive or capital-intensive?

Solution

$$MRTS = \frac{MPL}{MPK} = \frac{0.5 \cdot 2L^{-0.5}K^{0.5}}{0.5 \cdot 2L^{0.5}K^{-0.5}} = \frac{K}{L}$$

Optimality condition:

$$\frac{K}{L} = \frac{w}{r} = \frac{8}{2} = 4 \quad \Rightarrow \quad K = 4L$$

Constraint ($\alpha + \beta = 1$):

$$40 = 2L^{0.5}(4L)^{0.5} = 2L^{0.5} \cdot 2L^{0.5} = 4L \quad \Rightarrow \quad L^* = 10, \quad K^* = 40$$

$$C^* = 8 \cdot 10 + 2 \cdot 40 = 80 + 80 = 160$$

Capital-intensive: $K^* = 40 \gg L^* = 10$ because capital costs much less ($r = 2 \ll w = 8$). The firm almost completely replaces labor.

Exercise: Comparative statics

Starting from the previous exercise ($Q = 2L^{0.5}K^{0.5}$, $w = 8$, $r = 2$, $Q = 40$):

1. If w falls to 2 (same r), how does K/L change?
2. Does the firm become more labor-intensive or capital-intensive?
3. Compute the new minimum cost C^{**} .
4. Compare C^{**} with $C^* = 160$: what do you notice?

Solution

$$\frac{K}{L} = \frac{w}{r} = \frac{2}{2} = 1 \quad \Rightarrow \quad K = L$$

$$40 = 2L^{0.5}L^{0.5} = 2L \quad \Rightarrow \quad L^{**} = 20, \quad K^{**} = 20$$

$$C^{**} = 2 \cdot 20 + 2 \cdot 20 = 40 + 40 = 80$$

With $w = 2$ (**equal to** r): the firm is perfectly balanced ($K = L$), now labor-intensive relative to the previous case.

$C^{**} = 80 < C^* = 160$: the cost has halved. With a much lower w , the same output costs much less.

Economic interpretation

Low-wage countries (high L , low w) tend to attract labor-intensive production. High- w countries invest in automation to compensate.

Overall summary

- Problem: $\min wL + rK$ s.t. $f(L, K) \geq Q$
- Isocost: intercepts and slope $-w/r$
- Optimum: $MRTS = w/r$ or $\frac{MPL}{w} = \frac{MPK}{r}$
- Cobb–Douglas: $K/L = (\beta/\alpha)(w/r)$, then the constraint Q
- Comparative statics: $w \uparrow \Rightarrow K/L \uparrow$

Skills acquired

- ✓ Construct and read isocosts
- ✓ Check the optimality condition
- ✓ Solve Cobb–Douglas cost minimization
- ✓ Do comparative statics on w and r
- ✓ Read business choices (automation, offshoring) as optimization

Optional exercises for home

1. With $TC = 50 + Q^2$: compute AC , AVC , MC and the optimal Q^* .
2. With $Q = L^{0.4}K^{0.6}$, $w = 5$, $r = 10$, $Q = 20$: find L^* , K^* , C^* .